

Unit II

DATA-LINK LAYER & MEDIA ACCESS

Course Objective: COB2 - To know functions of Data link layer and Media access layer.

Course Outcome: CO2 - Comprehend the data flow and control between nodes in the network

Bloom's Taxonomy Level: K1, K2, K3

PartA

S.No.	Question	BTL*
1.	State the main services provided by the Data Link Layer. The data link layer provides four key services: <i>framing</i> for dividing the bit stream into frames, <i>error control</i> to detect and correct transmission errors, <i>flow control</i> to prevent sender–receiver speed mismatch, and <i>media access control</i> to regulate how devices share the transmission medium.	K1
2.	Mention the purpose of framing in the Data Link Layer. Framing is the process of dividing the continuous bit stream from the network layer into discrete, manageable units called frames. Each frame contains a header and trailer for identification and error detection. This helps in reliable data delivery between nodes.	K1
3.	Distinguish between FDMA and TDMA based on their method of channel allocation. FDMA assigns each user a distinct frequency channel, allowing simultaneous transmissions on different frequencies. TDMA allocates different time slots to each user on the same frequency channel. Both are channelization methods in media access control.	K2
4.	Describe how CSMA/CD helps in handling collisions during data transmission. CSMA/CD listens to the medium before sending data and detects collisions during transmission. If a collision occurs, the sender stops immediately, waits for a random backoff time, and retransmits. This reduces repeated collisions in Ethernet.	K2
5.	Specify the minimum and maximum frame sizes defined by IEEE 802.3 Ethernet. Ethernet frames have a minimum size of 64 bytes to ensure collision detection and a maximum size of 1518 bytes to limit transmission delay. These limits help maintain efficiency and compatibility in network communication.	K1
6.	Describe how ARP is used to obtain the MAC address corresponding to a known IP address. The Address Resolution Protocol (ARP) maps a known IP address to its corresponding MAC (physical) address within a local network. When a device needs to send data to another device on the same LAN, it uses ARP to discover the destination's hardware address. It operates at the interface between the network layer (IP) and the data link layer (MAC) .	K2
7.	Name any two error detection methods used in data communication. Two widely used error detection methods are cyclic redundancy check (CRC) and checksum, both used to detect changes to raw data during transmission. 1. Cyclic Redundancy Check (CRC) – Uses polynomial division to detect transmission errors. 2. Checksum – Sums all data units and compares the result with a transmitted checksum to identify errors	K1
8.	Describe the role of flow control in ensuring reliable data transmission between a sender and receiver. Flow control is a technique in the data link layer that manages the pace of data transmission between sender and receiver. It ensures that the sender does not transmit data faster than the receiver can process it. This prevents buffer overflow at the receiver's end. By balancing the transmission rate, it avoids data loss and ensures reliable communication. Common flow control methods include stop-and-wait and sliding window protocol.	K2

9.	List any two fields in the IEEE 802.3 frame format. Two important fields are: 1. Destination Address – Holds the MAC address of the receiver. 2. Source Address – Holds the MAC address of the sender.	K1
10.	Describe the primary applications of Bluetooth technology in short-range wireless communication. Bluetooth is a short-range wireless communication technology designed to connect devices without using physical cables. It operates in the 2.4 GHz ISM band and uses low-power radio waves, making it ideal for battery-operated devices. Its primary use is to transfer data or stream audio between compatible devices within a limited range, typically up to 10 meters for Class 2 devices. Common applications include wireless headphones, keyboards, mice, file sharing between phones, and connecting IoT gadgets.	K2
11.	Summarize the role of ICMP in error reporting and network diagnostics. The Internet Control Message Protocol (ICMP) is a network layer protocol used for sending control messages, error reports, and network diagnostics between devices. It helps in reporting issues such as unreachable hosts, network congestion, and timeouts, and is also used by tools like ping and traceroute for troubleshooting.	K2
12.	State the purpose of Network Address Translation (NAT). Network Address Translation (NAT) is a technique used by routers to modify IP address information in packet headers while forwarding traffic. It allows multiple devices in a private network to share a single public IP address, improving security and conserving IPv4 addresses. Common types include Static NAT, Dynamic NAT, and PAT (Port Address Translation).	K1
13.	What information does RARP provide to a device when its MAC address is known? The Reverse Address Resolution Protocol (RARP) is used by a device to obtain its IP address from a known MAC address. It is often used by diskless workstations during booting, where the device sends a request to a RARP server to get its IP address. RARP works in the opposite direction of ARP, enabling hardware-based identification of devices on a network.	K1
14.	State the purpose of a MAC address in a local area network. A MAC (Media Access Control) address is a unique hardware identifier assigned to every network interface card (NIC). It is 48 bits long, written in hexadecimal (e.g., 00:1A:2B:3C:4D:5E), and is used by the Data Link Layer for identifying devices in the same LAN. Unlike an IP address, a MAC address is fixed and is not location-dependent. It ensures correct delivery of frames in Ethernet, Wi-Fi, and Bluetooth networks.	K1
15.	Differentiate between unicast, multicast, and broadcast in the Data Link Layer. • Unicast: Data is sent from one device to one specific device. Example: Sending a file from one computer to another via LAN. • Multicast: Data is sent to a selected group of devices. Example: Streaming video to multiple subscribed users on a network. • Broadcast: Data is sent to all devices in the network segment. Example: ARP request to find a MAC address. These transmission modes are identified by special MAC address formats in the Data Link Layer.	K2

16.	Describe the concept of piggybacking and state how it improves communication efficiency. Piggybacking is a technique used in bidirectional communication where acknowledgment (ACK) frames are combined with outgoing data frames instead of being sent separately. This reduces the number of frames transmitted, thus saving bandwidth and improving efficiency. Example: In the Sliding Window Protocol, when device A receives data from device B, it sends an acknowledgment along with its own outgoing data in the same frame.	K2
17.	Name any two protocols that operate in the Data Link Layer. Ethernet (IEEE 802.3): Widely used wired LAN protocol that defines frame format, addressing, and CSMA/CD for media access. Point-to-Point Protocol (PPP): Used for direct communication between two nodes, such as over serial cables, DSL, or VPNs. These protocols define how frames are formatted, addressed, transmitted, and error-checked at the Data Link Layer.	K1
18.	Mention the meaning of CDMA in media access control. Code Division Multiple Access (CDMA) allows multiple stations to transmit simultaneously over the same frequency band by assigning each station a unique code to modulate its signal. It is used in 3G mobile networks and satellite communication.	K1
19.	State how TDMA allocates the communication channel among multiple stations. Time Division Multiple Access (TDMA) divides the available channel time into fixed time slots, and each station is assigned a slot for transmission. This avoids collisions and is used in GSM mobile networks and satellite communication.	K1
20.	Mention how FDMA allocates the available bandwidth among multiple stations. Frequency Division Multiple Access (FDMA) divides the available bandwidth into separate frequency bands, and each station is assigned a specific band for transmission. This method is used in traditional radio broadcasting, analog cellular systems, and some satellite links.	K1

PartB

S.No.	Question	BTL*
1	Describe the services provided by the Data Link Layer and discuss their role in reliable data communication. (16m)	K2
2	Compare the different media access control techniques with suitable diagrams and examples. (16m)	K2
3	Summarize the IEEE 802.3 Ethernet standard, including its frame structure and key features. (16m)	K2
4	Discuss the architecture, operation, and key features of IEEE 802.11 Wireless LAN. (16m)	K2
5	Illustrate the working of Bluetooth technology and discuss its applications in wireless communication. (16m)	K2
6	A network experiences slow communication, unreachable devices, and IP-to-MAC address resolution problems. Analyze how ICMP, NAT, ARP, and RARP can be used to address these issues with suitable examples. (16m)	K3
7	Compare the different error detection techniques used in data communication with suitable examples, advantages, and applications. (16m)	K2
8	Compare the different flow control methods used in data communication with respect to their operation, efficiency, and applications. (16m)	K2